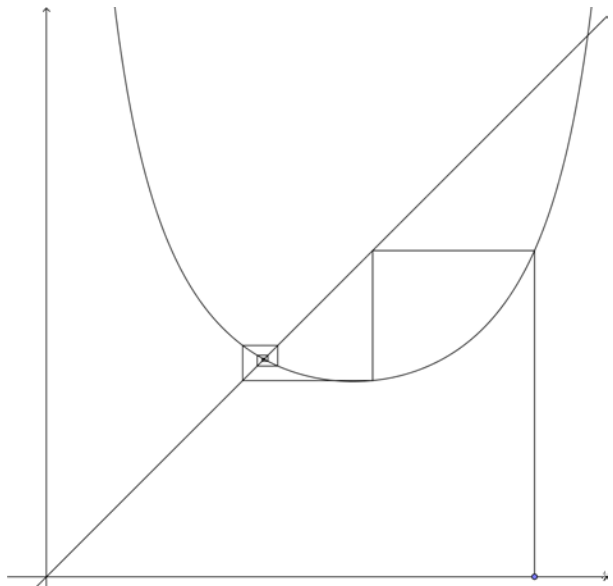
5	**(a)**	**(i)**	$x = 0$ $x = \pi$	**B1** **B1** **[2]**	**1.1** **1.1**	 180 gets 0	If answers given in both degrees and radians follow inst 2g
5	**(a)**	**(ii)**	$\left(\frac{\pi}{2},\ \ 1\right)$	**B1** **[1]**	**2.2a**	(90, 1) or (1.57, 1) get 0	
5	**(b)**		$1 - \text{cosec}1 = -0.188...$ or 'negative' $2 - \text{cosec}\ 2 = 0.900$ or 'positive' Change of sign so root between 1 and 2	**B1** **E1** **[2]**	**1.1a** **2.4**	Both correct Condone no mention of continuity AG	OE E,g, may use $\text{cosec}x - x$ Dep on B mark
5	**(c)**	**(i)**	**BC** 1.18840…, 1.07785…	**B1** **[1]**	**1.1a**	Both correct to at least 3dp.	
5	**(c)**	**(ii)**	**BC** 1.114	**B1** **[1]**	**2.2a**		
5	**(d)**		No, it converges to 1.114	**E1** **[1]**	**1.1**	OR same as *their* **(c) (ii)** or 'the root between 1 and 2' etc	Just 'No' gets 0 'Yes' with anything gets 0

Question			Answer	Marks	AOs		Guidance
5	(a)	(i)	$x = 0$	B1	1.1		If answers given in both degrees and radians follow inst 2g
5	(e)			B1 B1 B1 [3]	3.2a 2.2a 1.1	Starting point between min and right asymptote Initial "staircase" (≥2 horiz sections) Spirals into lower root	3 B marks all independent)